

Answer: B

$$11 \quad \omega = \frac{2\pi}{T} = \frac{2\pi}{365 \times 24 \times 3600} = 1.99 \times 10^{-7} \text{ rad s}^{-1}.$$

$$v = \omega r = \frac{2\pi}{365 \times 24 \times 3600} \times 1.50 \times 10^{11} = 2.99 \times 10^4 \text{ m s}^{-1}.$$

Answer: A

12

